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Insights into process safety incidents from an analysis of CSB investigations

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ABSTRACT

Many incidents have helped to define and develop process safety. Each has provided valuable learning opportunities. However, it is even more important to identify insights that can be obtained from an analysis of a large set of incidents that represents those that typically occur. This larger picture illuminates trends and commonalities and provides learning opportunities that are even more important than the causes of any one individual incident.

The Chemical Safety Board has published the results of over 60 investigations of process safety incidents. These data have been analyzed to identify commonalities and trends so that measures to help protect against future incidents can be developed. Recommendations are made to address key issues identified.

1. Introduction

Process safety regulations have been promulgated in various parts of the world to address protecting process plants against catastrophic incidents in response to the occurrence of a number of major incidents such as Flixborough, Seveso, and Bhopal (Atherton and Gil, 2008). However, despite the widespread adoption of process safety practices, major incidents continue to occur. The continued occurrence of such incidents is unacceptable and the issue of what can be done to protect against them must be addressed. This paper addresses the issue of why such incidents continue to occur.

Process safety regulations contain requirements for periodic audits for programs to identify any omissions or deficiencies and some process safety programs require periodic management reviews. However, while omissions and deficiencies must be identified and corrected, knowledge of what omissions and deficiencies have actually resulted in process safety incidents is essential to help prevent further incidents.

Compilations of incident descriptions have been published by various authors, for example, references (Gugan, 1979; Kletz, 2001, 2003, 2009; Atherton and Gil, 2008; M&M, 2014; Sanders, 2015). Usually, incidents are organized in various ways, such as by root cause, release type, equipment involved, operation being conducted, and process safety element involved. Descriptions of individual incidents also can be found, for example, in reports from the U.S. Chemical Safety Board (CSB) (CSB), and in journals such as the Loss Prevention Bulletin (LPB). Such publications usually focus on providing a description of the incident, identifying its causes, and providing recommendations for corrective actions. Certainly, lessons can be learned from each incident. However, it is also important to take an overall view to identify commonalities and trends across incidents so that measures can be identified that will help to protect against future incidents. In particular, the identification of issues that have contributed to multiple incidents is especially important so that future incidents may be prevented. For example, the CSB has flagged reactive hazards (CSB, 2002), combustible dusts (CSB, 2006), and hot work (CSB, 2010) as significant issues across multiple incidents. This paper identifies additional issues that were important contributors in multiple incidents.

CSB investigation reports have been analyzed concerning management system failures (Blair, 2004), process hazard analysis (PHA) (Kaszniak, 2010), and the hierarchy of controls (Amyotte et al., 2011). Also, the evolution of process safety standards and legislation following nine key process safety incidents has been addressed (Kerin). This paper analyzes incidents investigated by the CSB to identify trends and commonalities that have not yet been identified or analyzed. This knowledge can be used to help prevent future incidents.

Section 2 identifies the incidents reviewed and the analysis

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approach followed. Section 3 identifies and discusses the common contributors that were found for the incidents studied and provides recommendations to address them. Section 4 provides a discussion of the results. Section 5 provides some additional noteworthy observations that were made during the review of incident descriptions. Section 6 identifies and discusses three issues that underlie all of the other issues. Section 7 provides recommendations to strengthen process safety programs. Conclusions are presented in section 8.

2. Incidents addressed and analysis approach

Incidents investigated by the CSB were reviewed (CSB). The CSB has issued over 60 incident investigation reports, case studies, and safety bulletins dating back to 1998 when the CSB began investigating incidents. The incidents cover various sectors of the process industries. The CSB currently is investigating one incident from 2010, three from 2013, three from 2014 and one from 2015. Sixty-eight CSB incidents were studied (see Table 1). About 40% of the CSB incidents occurred in processes covered by the Occupational Safety and Health Administration's (OSHA's) process safety management (PSM) regulations. The descriptions of incidents and their causes were taken at face value.

All incident descriptions were first reviewed to identify commonalities. A second reading of the incident descriptions was performed to confirm the initial findings and document characteristics of the incidents. A third reading of the incident descriptions was conducted to corroborate the findings and provide further detail.

3. Analysis of incidents

This section identifies common contributors to the incidents. Of course, each incident has its own individual characteristics but there is a remarkable degree of similarity in some deficiencies and omissions across the incidents. The focus in this paper is on issues common to many of the incidents.

Incidents where each issue was a contributor are identified, the percentage of CSB incidents where the issue was present is given, aspects of each issue are discussed, and recommendations are made to address the issues.

3.1. Process design

Many incidents occurred in processes that arguably had deficient designs {1, 3, 6, 15, 16, 17, 18, 21, 25, 29, 31, 34, 37, 40, 41, 42, 57, 58, 64}[1]. This issue was present in 28% of CSB incidents and 18% of incidents covered by OSHA's PSM standard. Deficiencies arose in the initial design and as a result of modifications. Hazards were not recognized, designs created hazards, safety systems were inadequate, process instrumentation was insufficient, human factors issues were not addressed, hazards of scale-up were not recognized, and throughput was increased without review. In some cases, deficient equipment design contributed to incidents {54a, 54c}. Possibly designers were unaware of the problems and/or design reviews were inadequate but these reasons are not acceptable.

Design deficiencies can be addressed by making process design and engineering an element in process safety programs (Baybutt, 2016). Examples of topics that should be addressed by requirements for process design and engineering include design practices, design reviews, inherent safety, hazardous area classification, pressure relief, alarm management, damage mechanisms, human factors, facility siting, competency, and training for designers. Also, process design and engineering should be covered by the same management system requirements as other elements in a process safety program such as written policies and procedures, document management, assigning roles and responsibilities, auditing, management review, and continuous improvement. Process design and engineering should be an element of process safety management systems in order to provide a specific focus on the activities involved and to ensure the application of management system practices to it. Also, the competency of process designers must be addressed.

For new designs and process modifications, proof of safety should be required rather than proof of danger. This will require a change in culture.

3.2. Safeguards

Many incidents involved processes that were not adequately protected by safeguards {2, 7, 8, 9, 11, 12, 13, 17,18, 20, 21, 22, 24, 25, 26, 28, 29, 30, 32, 34, 35, 36, 37, 38, 40, 41, 42, 46, 48, 49, 50, 51, 54a, 54b, 54c, 60, 62, 64}. This issue was present in 56% of CSB incidents and 57% of incidents covered by OSHA's PSM standard. Needed safeguards were not present or those that were present were insufficient, unreliable, disabled, or lacked independence. Commonly, the hierarchy of controls was not applied (see Fig. 1). Often reliance was placed on procedural safeguards which are at the lower end of the hierarchy (19a, 22, 25, 29, 32, 46, 54c}. Process risks should be matched with safeguards in the hierarchy of controls. Higher consequence scenarios, such as a chlorine release in a densely populated area, require safeguards higher in the hierarchy and likely multiple, independent safeguards. Procedural safeguards typically are viewed as less reliable, and, therefore, inadequate for protection against high-consequence events.

Inherently safer designs or practices could have prevented or mitigated some incidents {29, 39, 55, 56, 54b, 63}. Currently, there are no requirements to address inherently safer systems or the hierarchy of controls in Federal process safety regulations. Process safety regulations in New Jersey require an inherently safer technology review (NJDEP) and in Contra Costa County, California require that inherently safer systems be addressed (CCHS). Proposed amendments to California's process safety regulations for refineries contain a requirement to perform a hierarchy of controls analysis (DIR, 2015).

The operability of safeguards was an issue during several incidents. Explosions and fires disabled safeguards {15, 30, 42}, and toxic releases and fires prevented access to controls {19a, 30, 37, 50}.

Currently, there is no requirement in US process safety regulations to address the design basis for safeguards, evaluate or document the effectiveness of process safeguards, or show that safeguards adequately reduce risks. Proposed amendments to California's process safety regulations for refineries contain a requirement to perform a hierarchy of controls analysis and a safeguards protection analysis (SPA) (DIR, 2015). The SPA should be performed before a PHA study and updated after the PHA study has been completed. The SPA should include the preparation of a Safety Requirements Specification (SRS) for each individual safeguard (Baybutt, 2013b). An SRS documents the functional and integrity requirements for a safeguard. Functional requirements describe what the safeguard does to prevent a hazardous state of the process and integrity requirements specify how well the safeguard should perform and what is needed to support that performance level. The SPA should validate that safeguards adequately protect against the hazards of the process, have a specified integrity level, have been

[1] Braces or curly brackets, that is, {}, are used to enclose the number designations of incidents from Table 1.

Table 1
CSB incidents reviewed.

No.	Incident	Incident date	Report date	PSM[1]	PHA[2]
1	Sonat Exploration Co. Catastrophic Vessel Overpressurization	03/04/1998	09/21/2000	N	N
2	Morton International Inc. Runaway Chemical Reaction, Explosion and Fire	04/08/1998	08/16/2000	N	Y
3	Concept Sciences Hydroxylamine Explosion	02/19/1999	02/01/2002	Y	Y
4	Tosco Avon Refinery Petroleum Naphtha Fire	02/23/1999	03/21/2001	Y	N
5	Bethlehem Steel Corporation Gas Condensate Fire	02/02/2001	12/06/2001	N	N
6	BP Amoco Thermal Decomposition Incident	03/13/2001	05/20/2002	N	Y
7	Motiva Enterprises Sulfuric Acid Tank Explosion	07/17/2001	08/28/2002	N	Y
8	Georgia-Pacific Corp. Hydrogen Sulfide Leak	01/16/2002	11/20/2002	N	N
9	Third Coast Industries Petroleum Products Facility Fire	05/01/2002	03/06/2003	N	U
10	DPC Enterprises Festus Chlorine Release	08/14/2002	05/01/2003	Y	U
11	First Chemical Corp. Reactive Chemical Explosion and Fire	10/13/2002	10/15/2003	N	N
12	Environmental Enterprises Hydrogen Sulfide Release	12/11/2002	09/17/2003	N	U
13	Catalyst Systems Inc. Reactive Chemical Explosion and Fire	01/02/2003	10/29/2003	N	N
14	BLSR Operating Ltd. Vapor Cloud Deflagration and Fire	01/13/2003	09/17/2003	N	U
15	West Pharmaceutical Services Dust Explosion and Fire	01/29/2003	09/23/2004	N	N
16	Technic Inc. Ventilation System Explosion and Fire	02/07/2003	08/20/2004	N	N
17	CTA Acoustics Dust Explosion and Fire	02/20/2003	02/15/2005	N	N
18	D.D. Williamson & Co. Catastrophic Vessel Failure	04/11/2003	03/12/2004	N	N
19	Honeywell Chemical Releases (3 incidents)	a. Chlorine 07/20/2003	08/08/2005	Y	Y
		b. Antimony pentachloride 07/29/2003		N	N
		c. Hydrogen fluoride 08/13/2003		Y	N
20	Isotec/Sigma Aldrich Nitric Oxide Explosion	09/21/2003	08/24/2004	Y	Y
21	Hayes Lemmerz Dust Explosions and Fire	10/29/2003	09/27/2005	N	N
22	DPC Enterprises Glendale Chlorine Release	11/17/2003	02/28/2007	Y	Y
23	Giant Industries Oil Refinery Explosion and Fire	04/8/2004	10/26/2005	Y	Y
24	MFG Chemical Inc. Toxic Gas Release	04/12/2004	04/11/2006	Y	N
25	Formosa Plastics Vinyl Chloride Explosion	04/23/2004	03/06/2007	Y	Y
26	Sterigenics Ethylene Oxide Explosion	08/19/2004	03/30/2006	Y	Y
27	Marcus Oil and Chemical Tank Explosion and Fire	12/03/2004	06/06/2006	N	N
28	Acetylene Service Company Gas Explosion	01/25/2005	01/26/2006	Y	Y
29	BP Texas City Refinery Explosion and Fire	03/23/2005	03/20/2007	Y	Y
30	Formosa Plastics Propylene Fire and Explosions	10/06/2005	07/20/2006	Y	Y
31	Bethune Point Wastewater Plant Explosion and Fire	01/11/2006	03/13/2007	N	N
32	Synthron Chemical Reaction and Vapor Cloud Explosion	01/31/2006	07/31/2007	Y	N
33	Partridge Raleigh Oilfield Explosion and Fire	06/05/2006	06/12/2007	N	N
34	Universal Form Clamp Co. Explosion and Fire	06/14/2006	04/10/2007	Y	N
35	EQ Hazardous Waste Plant Explosions and Fire	10/05/2006	04/16/2008	N	U
36	CAI/Arnel Chemical Plant Explosion	11/22/2006	05/13/2008	Y	N
37	Valero Refinery Propane Fire	02/16/2007	07/09/2008	Y	Y
38	Barton Solvents Explosions and Fire	07/17/2007	06/26/2008	N	U
39	Xcel Energy Company Hydroelectric Tunnel Fire	10/02/2007	08/25/2010	N	Y
40	Barton Solvents Flammable Liquid Explosion and Fire	10/29/2007	09/18/2008	N	U
41	T2 Laboratories Inc. Reactive Chemical Explosion	12/19/2007	09/15/2009	N	N
42	Imperial Sugar Company Dust Explosion and Fire	02/07/2008	09/24/2009	N	U
43	Goodyear Heat Exchanger Rupture and Ammonia Release	06/11/2008	01/27/2011	Y	U
44	Packaging Corporation Storage Tank Explosion	07/29/2008	03/04/2010	N	N
45	Bayer CropScience Pesticide Waste Tank Explosion	08/28/2008	01/20/2011	Y	Y
46	INDSPEC Chemical Corporation Oleum Release	10/11/2008	09/30/2009	Y	Y
47	Allied Terminals Fertilizer Tank Collapse	11/11/2008	05/26/2009	N	N
48	Veolia Environmental Services Flammable Vapor Explosion and Fire	05/04/2009	07/21/2010	Y	N
49	ConAgra Natural Gas Explosion	06/09/2009	02/04/2010	N	U
50	CITGO Refinery Fire and Hydrofluoric Acid Release	07/19/2009	12/09/2009	Y	Y
51	Caribbean Petroleum Refining Tank Explosion and Fire	10/23/2009	10/21/2015	N	N
52	Silver Eagle Refinery Explosion	11/04/2009	04/14/2014	Y	U
53	NDK Crystal Inc. Explosion	12/07/2009	11/14/2013	N	U
54	DuPont Corporation Toxic Chemical Releases (3 incidents)	a. Methyl chloride 01/22/2010	09/20/2011	U	U
		b. Oleum 01/23/2010		U	U
		c. Phosgene 01/23/2010		Y	Y
55	Kleen Energy Natural Gas Explosion	02/07/2010	06/28/2010	N	U
56	Tesoro Refinery Explosion and Fire	04/02/2010	05/01/2014	Y	Y
57	Horsehead Holding Company Explosion and Fire	07/22/2010	03/11/2015	N	U
58	Millard Refrigerated Services Ammonia Release	08/23/2010	01/15/2015	U	U
59	E. I. DuPont De Nemours Co. Hotwork Explosion	11/09/2010	04/19/2012	Y	Y
60	AL Solutions Dust Explosion	12/09/2010	07/16/2014	N	U
61	Carbide Industries Fire and Explosion	03/21/2011	02/07/2013	N	U
62	Hoeganaes Corporation Flash Fires and Explosion	05/27/2011	01/05/2012	U	U
63	Chevron Refinery Fire	08/06/2012	01/28/2015	Y	Y
64	US Ink Fire	10/09/2012	01/15/2015	U	U

Y – Yes.
N – No.
U – Unknown.
[1] Covered by OSHA's PSM standard.
[2] PHA was performed.

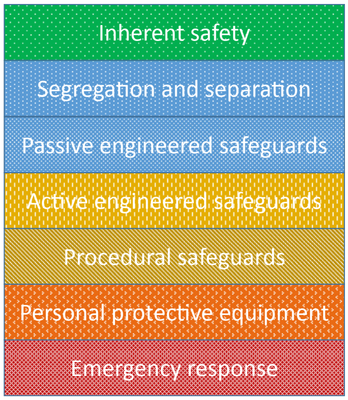

Fig. 1. Hierarchy of controls.

installed correctly, are functional, are being maintained appropriately, etc.

Once the SPA has been performed, the PHA team can focus on the qualification of safeguards for specific hazard scenarios (Baybutt, 2012b, 2013b). Qualification of safeguards addresses whether safeguards are appropriate for each hazard scenario in a PHA study. Information developed in a PHA study may necessitate revisions to the SPA and SRSs. For example, it may be determined that a redundant power supply is needed to support the integrity required for a scrubber system. SPAs should be updated during management of change (MOC) reviews and when developing recommendations from incident investigations.

3.3. Operations and maintenance

Problems with procedures were common {1, 2, 3, 5, 6, 11, 12, 13, 14, 18, 19b, 19c, 21, 22, 25, 28, 29, 32, 33, 36, 41, 42, 45, 46, 47, 51, 54c, 59, 61, 62}. This issue was present in 44% of CSB incidents and 43% of incidents covered by OSHA's PSM standard. In numerous incidents, procedures had not been developed, were not followed, were incomplete, or were otherwise deficient. Such situations are unacceptable. Properly written procedures must be developed following best practices for all process operations and activities before they occur and for all stages and modes of operation. Procedures are needed to deal with any situation where there is the potential for a process safety incident. Personnel need to be trained in the procedures and practice them regularly in drills. Facility personnel must be competent.

In six CSB investigations that identified problems with standard operating procedures (SOPs), the CSB noted that each incident involved SOP deviations that became “necessary violations” to get the job done {45}. Such situations cannot be tolerated.

Existing process safety regulations require written operating procedures. However, the regulations should be clarified to encompass all process tasks and activities and a mechanism is needed to ensure the regulatory requirements are followed.

Deficiencies in maintenance were present in a number of incidents {5, 12, 21, 29, 31, 32, 51, 54c}. Issues regarding the adequacy of inspections were common {4, 7, 47, 53, 56, 62}. For some incidents, there was no or an inadequate preventive maintenance (PM) program {11, 13, 16, 41} and no or inadequate mechanical integrity (MI) program for some incidents {10, 23, 29, 52, 61, 63}. Such issues were present in 34% of CSB incidents and 32% of incidents covered by OSHA's PSM standard. Existing process safety regulations address the integrity of equipment but a mechanism is needed to ensure the requirements are followed for all safety-critical equipment. Current US process safety regulations require that covered equipment comply with recognized and generally accepted good engineering practices (RAGAGEP) but the term is not defined. OSHA issued a Request for Information (RFI) in advance of rulemaking to revise the PSM standard (FR, 2013). Amongst other topics, the RFI sought comment on whether an explicit definition of RAGAGEP should be provided, and, if so, what definition should be used. The current OSHA PSM regulation provides a limited list of covered equipment (CFR) and the RFI sought comments on whether mechanical integrity requirements should apply to all equipment designated as safety-critical equipment.

The Risk Based Process Safety (RBPS) program of the Center for Chemical Process Safety (CCPS) contains the element, Conduct of Operations, which provides requirements for operational discipline that tolerate no deviation from approved procedures, even if the outcome is inconsequential or desirable (AIChE, 2007b, 2011). Appropriate work procedures must be determined, documented, and enforced. The importance of following procedures must be emphasized to facility personnel. The adoption of an element on operational discipline would help in this regard coupled with an element that addresses process safety culture. The CCPS RBPS program contains both these elements but they are not included in existing US regulations.

An operational discipline element would encourage and enforce strict adherence to established procedures and imbue facility personnel with the belief that no operation or activity should be performed without a governing procedure. When people's lives are at risk it should not be acceptable to perform activities without proper preparation or by improvising.

Another deficiency present in several incidents was the acceptance of practices as normal and acceptable by facility personnel that outside observers would view as deviant and unacceptable {17, 19c, 21, 29, 42, 57, 62, 63}. This phenomenon has been called “normalization of deviance” (Vaughn, 1996). It involves allowing operation of a process that deviates from established limits by relaxing performance standards rather than identifying and correcting the underlying causes of deviations so the deviations become the new, normal operating conditions. Hazards have been normalized in some cases {12, 27, 46, 56}. Normalization of deviance was present in 18% of CSB incidents and 18% of incidents covered by OSHA's PSM standard.

While the acceptance of a lesser performance standard in such cases may not immediately result in an incident, continued successful operation instills a sense of false security. It will be only a matter of time before uncorrected hazardous conditions eventually result in an incident. Unfortunately, facility personnel can become so accustomed to exposure to hazardous situations that the risks are not recognized even when their lives are at stake. Just because a facility has not experienced a catastrophic incident, perhaps ever in its lifetime, does not mean one could not happen tomorrow.

Normalization of deviance can be addressed by ensuring that any deviations from established performance standards are identified and resolved as soon as they occur. Operational discipline will help in this regard.

3.4. Abnormal and non-routine operations

Many incidents involved non-routine operations and situations {1, 3, 4, 5, 6, 10, 11, 16, 17, 19c, 20, 21, 23, 24, 26, 27, 29, 32, 33, 39, 41, 43, 45, 46, 47, 49, 52, 54a, 54c, 55, 56, 58, 59, 60, 61, 62, 63, 64}. In some cases, they were not recognized as non-routine operations and were not reviewed to ensure that work could proceed safely. Flaws in hot work {7, 31, 33, 44, 59} and other safe work practices {4, 5, 6, 11, 14, 23, 39, 43} also contributed to incidents. Lack of equipment isolation was a contributor to some incidents {11, 33, 59}. Abnormal and non-routine operations were involved in 62% of

CSB incidents and 64% of incidents covered by OSHA's PSM standard.

People play an intimate role in non-routine operations. This both increases the likelihood of human error being a factor and places people closer to process hazards. Furthermore, a number of incidents occurred in processes that involved manual operations or were manually controlled {2, 12, 13, 16, 18}.

Current US process safety regulations contain requirements to develop and implement safe work practices and utilize hot work permits. However, the regulations do not specifically address non-routine operations. There are requirements for written procedures under the Operating Procedures element for specific phases of operation listed in the regulations (initial startup, normal operations, temporary operations, emergency shutdown, emergency operations, normal shutdown, and startup following a turnaround or after an emergency shutdown). However, other phases of operation are not specified including commissioning, standby, bypass, unattended operation, maintenance on live equipment, regeneration, troubleshooting, operation with disabled safety systems, cleaning, decommissioning, etc. Process safety programs must address all phases of operation that are or may be experienced by a process. Emphasis should be placed on non-steady state operations including abnormal operations because they are the process states in which most incidents have occurred.

Other elements of process safety programs besides operating procedures should include requirements to address non-routine operations, for example, mechanical integrity and PHA (Baybutt, 2012a). A separate element on non-routine operations may be merited to provide requirements in a single place. Absent specific requirements to address non-routine operations, many non-routine operations are likely not to be viewed as such and their hazards not recognized or addressed.

Procedures are needed for identifying and planning for non-routine work and abnormal situations, e.g. restart after emergency shutdown or an extended unplanned shutdown. Processes should be operated defensively and conservatively. Procedures should specify alternative courses of action if safety preconditions cannot be met. Furthermore, job hazard analysis (Swartz, 2001) should be performed immediately prior to any non-routine activity.

3.5. Human and organizational factors

Human and organizational factors contributed to many incidents, for example, valve positions were not clear {23, 51}, valve operation was difficult {56}, lighting was insufficient {51}, operators were distracted {18}, operators were fatigued {29, 45}, operators had insufficient information {57}, procedures were complex and lengthy {45}, communications were problematic {25, 29, 43, 45}, layout was error prone {25}, staffing was not sufficient {25, 29, 45, 51}, operators became desensitized to an alarm due to its history of unreliability {54a}, the work environment encouraged operations personnel to deviate from procedures {29}, a computerized control system was poorly designed {29}, a new control system posed challenges {45}, and there were difficulties with organizational structure {25, 29}. Undoubtedly, problems with safety culture played an important role in many incidents, with BP Texas City being the best known {29}. Most CSB investigations have not explicitly identified contributions to incidents from human and organizational factors so it is not possible to provide meaningful statistics for their contributions to incidents. Undoubtedly, human and organizational factors played a role in most accidents.

Fundamentally, all of the incidents reviewed resulted from human failures ranging from design to operation to maintenance. Even when equipment failures such as corrosion were the basic cause of an incident, the underlying cause usually was human failure {63}.

Some incidents involved conflicts and tradeoffs between process safety and other imperatives {4, 7, 29, 39, 45, 51, 54c, 58, 63}, often financial but sometimes other issues such as noise attenuation {36}.

OSHA's PSM standard requires that "the process hazard analysis shall address human factors" (CFR). However, it is not sufficient simply to require PHA studies to address human factors. Certainly, PHA teams must address the role of human factors in influencing human failures but this approach leads to insufficient consideration of human factors issues for processes. Commonly, human factors are addressed by using a checklist with questions that are asked about the entire process so that analysts can try and think of any problems. A much more detailed approach is needed. Otherwise, many aspects of human factors will be missed because of practical limits on the level of detail that can be addressed in PHA studies, limitations on the ability of PHA methods to address human factors, and time pressures on PHA teams. Also, human factors issues affect each of the elements of a process safety program including PHA itself (Baybutt, 2013a). It is not reasonable to expect a PHA team to address human factors issues that impact its own performance. The Contra Costa industrial safety ordinance requires development and implementation of a written human factors program (CCHS). Proposed amendments to California's process safety regulations for refineries contain a requirement to develop, implement and maintain an effective written human factors program that goes beyond the requirements of Contra Costa's industrial safety ordinance (DIR, 2015).

All process safety management programs should have a distinct element that addresses human factors for processes with specific requirements to be addressed. Many aspects of human factors for process safety have been addressed by CCPS (AIChE, 2007a) and they should be covered by process safety programs.

For many incidents, it is difficult to believe that the conditions that resulted in the incident were allowed to occur. An independent observer easily can conclude that surely someone could have intervened at some stage in the sequence of events that occurred and taken action to prevent the incident. However, such a view does not take account of the psychological factors that influence decision making by people, particularly in groups and when under pressure to achieve competing goals. Factors such as group think, peer pressure, and other cognitive biases interfere with objective decision making (Baybutt, 2015a). Questionable decision making played a role in numerous incidents, for example {58, 63}. Facility personnel need to apply critical thinking in their daily work activities (Moore and Parker, 2015). Critical thinking can be taught and key facility personnel should be so trained.

Companies can benefit from integrating the characteristics of high reliability organizations into their operations such as mindful organizing for the unexpected (Weick and Sutcliffe, 2015).

3.6. Process hazard analysis

Failures in PHA are a common finding in CSB incident investigation reports. Indeed, issues with PHA were present for all CSB incidents where a PHA study was performed. Of particular note is that CSB found that incidents were not identified by the PHA studies performed. However, many of the incidents involved deliberate operation of processes in ways that a PHA team would not consider credible. PHA teams look for ways in which a process may deviate from its design intent but usually the specified design intent identifies what should be done, not what should *not* be done. It is the latter that occurs in many incidents and, while PHA teams try to identify ways that a process may be misoperated, it is unreasonable to expect a PHA team to identify the ways in which a process may be maloperated. Moreover, there is an essentially infinite number of ways in which a process may be maloperated and it would be impractical to ask a PHA team to try and identify all of them. Maloperation must be addressed through operating discipline, proper change reviews, and designing robust processes.

In some cases, the CSB found that PHA studies did not adequately address safeguards because safeguards listed in PHA studies were non-specific or general {19a, 56, 63} and, in many cases, relied too heavily on administrative procedures. For example, safeguards were recorded as "design", "standard operating procedures", "testing and inspection", and "proper emergency response procedures". PHA studies must identify specific safety-critical equipment, for example, a specific relief valve, and specific safety-critical actions, for example, activation of an emergency quench system by an operator. PHA teams should consider the hierarchy of controls when evaluating existing safeguards and deciding on recommendations for further risk reduction. PHA team leaders and team members must be competent (Baybutt, 2015b).

Some CSB incident investigation reports noted the failure to validate critical assumptions used in PHAs {45, 56, 29}. However, such validations should not be part of PHA. They should be performed before a PHA study is conducted (Baybutt, 2013b). Such validations require a different skill set than that for PHA team members. Moreover, if they were included in PHA studies they would add complexity to the studies and detract from their primary goal of scenario identification. Of course, the adequacy of safeguards for individual hazard scenarios must be addressed by the PHA team.

Some incidents have occurred when more than one activity was being performed simultaneously or sequentially and the activities conflicted {4, 7, 8, 25, 33, 43, 47, 48, 59}. Simultaneous and sequential operations (SIMOPS) reviews (IMCA, 2010) should be performed as adjuncts to PHA studies to address such issues. However, they are not yet common and there are no US regulatory requirements to perform them.

PHA studies were known to be performed for 22 of the 68 CSB incidents that were reviewed, although incident descriptions did not indicate whether or not a PHA study was performed for 21 of the incidents. However, it is likely that a PHA study was not performed for most of these incidents. Of the 22 PHA studies that were performed, only 4 studies were conducted for processes not covered by OSHA's PSM standard (see Table 1). This finding suggests that PHA studies are only performed when required by regulations.

3.7. Process changes

Many incidents involved changes that occurred but were not reviewed for their impact on process safety {2, 4, 6, 7, 8, 16, 17, 18, 21, 25, 27, 29, 37, 45, 46, 54c, 59, 63}, and inadequate reviews were performed in some cases {7, 15, 19a, 29, 32, 45, 54a, 56, 63}. Issues with process change reviews were present in 34% of CSB incidents and 39% of incidents covered by OSHA's PSM standard. Omissions and deficiencies can arise because changes are not recognized as requiring MOC reviews, personnel lack awareness of MOC requirements, or MOC reviews are bypassed for expediency. These problems relate to the competency of facility personnel and operational discipline. If personnel are involved in unauthorized and unreviewed changes, they cannot be viewed as competent to work in the facility in the same way as if they maloperated a process or performed maintenance work incorrectly. The establishment of a competency management program should help in this regard. It should be reinforced by operational discipline.

Also, MOC reviews required by the OSHA PSM standard have a narrow definition of covered changes. They cover only changes to process safety information (PSI), procedures, and facilities that affect a covered process (CFR). These requirements are neither global nor sufficiently specific and they are open to interpretation. Organizational changes have played a role in several incidents {29, 45, 56, 64} but they are not currently covered by the regulations, although OSHA's RFI addresses this omission (FR, 2013). However, incidents have involved other types of changes not explicitly defined in US process safety regulations including changes in feedstock {4, 63}, addition or removal of equipment {16, 22, 37, 42, 59}, degraded equipment {59}, scale-up of recipes {32}, and prolonged operation {6}. Furthermore, MOC programs should encompass operational deviations and variances, and abnormal situations as well as preplanned changes (CSB, 2001). Regulations and best practices should specify that any change that may impact on or affect a release of a highly hazardous chemical should be covered by MOC requirements.

Additionally, activities involved in making process changes can be hazardous and are not covered by current US process safety regulations. MOC reviews of changes should include the performance of job hazard analysis for the activities involved.

3.8. Previous incidents

Information from previous incidents that could have helped to prevent incidents was not used in many cases {2, 5, 6, 11, 17, 19a, 20, 21, 25, 26, 29, 41, 42, 45, 51, 53, 54a, 54c, 60, 61, 63, 64}. Issues with previous incidents were present for 32% of CSB incidents and 29% of incidents covered by OSHA's PSM standard. Previous incidents were not investigated properly, or at all, and near misses were ignored. The issue of retaining information from previous incidents in the corporate memory of companies is well known (Kletz, 1993). However, information cannot be retained if it has not first been developed.

Facility personnel may not have the time or inclination to study incident descriptions and they may not see or understand their relevance to their own facilities. Reliance cannot be placed on people to address lessons learned from previous incidents unless process safety programs contain specific requirements to do so. OSHA's PSM standard requires that previous incidents be addressed (CFR). However, the regulatory requirement is part of the PHA element which simply states that the PHA shall address previous incidents. This weak requirement is likely to result in the inadequate consideration of how information from previous incidents can be used to prevent repeat or similar incidents. Process safety programs should have a separate element on previous incidents, including near misses, that requires companies to identify relevant previous incidents from within the company and from other companies in the process industries, compile pertinent information on them, analyze the information in the context of reducing risks in existing processes, and develop recommendations to do so and track them to implementation. The previous incidents review should be performed periodically, e.g. annually, and documented. It should address both the issue of ensuring that incidents are identified, reviewed, and lessons learned are implemented, and that information is retained.

3.9. Facility and stationary source siting

The proximity of facilities to members of the public was an issue in multiple incidents {2, 3, 9, 10, 15, 16, 17, 18, 19a, 20, 22, 23, 24, 25, 26, 27, 29, 30, 32, 35, 36, 41, 45, 46, 48, 51, 52, 53, 63}. This issue was present for 43% of CSB incidents and 64% of incidents covered by OSHA's PSM standard. It raises concerns regarding land use planning in the US. Proximity of facility personnel to hazards in plants also was an issue in numerous incidents {11, 19a, 20, 21, 25, 26, 29, 32, 41, 48, 61}. This issue was present for 16% of CSB incidents and 25% of incidents covered by OSHA's PSM standard. The consequences of some incidents have been exacerbated by the presence of non-essential personnel {29, 49, 56}.

OSHA's PSM standard requires that "the process hazard analysis shall address facility siting" (CFR) and EPA's RMP rule requires that "the process hazard analysis shall address stationary source siting (FR, 1996). However, it is not sufficient simply to require PHA studies to address facility and stationary source siting. Certainly, PHA teams must address the impact of facility and stationary source siting on hazard scenarios but this approach likely leads to insufficient consideration of facility and stationary source siting issues for processes. Aspects of facility and stationary source siting will be missed because of practical limits on the level of detail that can be addressed in PHA studies, limitations on the ability of PHA methods to address facility and stationary source siting, and time pressures on PHA teams.

Process safety management programs should have a distinct element that addresses facility and stationary source siting for processes with specific requirements to be addressed. These requirements should include topics that often are overlooked including using designs that facilitate minimal operator presence; limiting the number of people exposed to hazards, particularly at more hazardous times by moving unnecessary people away from non-routine operations; training and requiring people to leave when hazardous conditions or situations arise; and assessing the potential for domino events.

The problem of the proximity of people to hazards is compounded by poor emergency preparedness and response which was present for about 40% of these incidents (see next section).

3.10. Emergency preparedness and response

Deficiencies in emergency preparedness and response existed in many cases {5, 8, 10, 11, 16, 19a, 20, 22, 24, 25, 32, 34, 35, 39, 42, 43, 45, 51, 54a, 54b, 63, 64}. This issue was present for 32% of CSB incidents and 39% of incidents covered by OSHA's PSM standard. Deficiencies occurred in hazard communication, evacuation procedures and emergency egress, community notification, emergency preparedness, knowledge of responders, and coordination between responders. OSHA's RFI addresses requiring facilities to coordinate emergency planning with local emergency response authorities (FR, 2013).

An effective way to help ensure that emergency response plans are in place and will work is to hold regular, periodic drills to test them. It is important to ensure that emergency preparedness and response correlates emergency response with actual hazard scenarios and addresses access to fire water monitors and HAZMAT equipment during incidents and provisions for containment of fire fighting water to prevent environmental pollution.

3.11. Audits and inspections

CSB investigations found that audits by companies, regulators, insurance providers, and others often did not identify process safety omissions and deficiencies and that audits and inspections missed issues that contributed to incidents in several cases {17, 18, 22, 42, 56, 59, 64}. Inadequate audits were identified in several cases {4, 22, 29}. Issues with audits and inspections were present for 13% of CSB incidents and 18% of incidents covered by OSHA's PSM standard.

The CSB noted that in the case of a refinery incident {56} a refinery National Emphasis Program (NEP) inspection was carried out about 18 months before the incident occurred but the mechanical integrity issues that were at the heart of the incident were not identified. NEP audits are the most comprehensive ones that have been performed by regulators. This audit was conducted by a team of people that spent several weeks at the refinery and it focused on the unit that subsequently experienced the catastrophic incident investigated by the CSB.

OSHA's PSM standard requires that a company conduct a compliance audit every three years to "verify that the procedures and practices developed under the standard are adequate and are being followed" (CFR). This terse requirement can make for perfunctory audits. Furthermore, process safety compliance audits do not focus on fundamental inadequacies in the design of a process safety program; rather they evaluate the design against regulatory requirements and their implementation. Audits may not properly address non-routine operations because usually they do not occur when audits are performed. Additionally, audits are conducted under time limitations and use sampling techniques.

More comprehensive audits should be performed. They should cover all aspects of a process safety program including, for example, safety culture, Auditors must be competent and should be certified (Baybutt, 2015d). OSHA's RFI addresses requiring third-party audits, increasing the required frequency of compliance audits, and requiring specific time frames for responding to deficiencies found by audits. There is an international standard for auditing management systems (ISO, 2011). Process safety regulations should require audits be conducted in compliance with the standard.

3.12. Equipment damage mechanisms

Equipment damage mechanisms played an important role in several incidents {4, 10, 19a, 20, 53, 56, 63}. Damage mechanisms are mechanical, chemical, physical or other processes that result in equipment or material degradation. They can lead to loss of process containment and process safety incidents.

Interestingly, all but one of these incidents occurred in a facility covered by OSHA's PSM standard. The CSB has noted the need to perform damage mechanism reviews (DMRs) {63}. Proposed amendments to California process safety regulations for refineries include requirements for conducting DMRs (DIR, 2015).

DMRs should be conducted as part of the mechanical or asset integrity element of process safety programs by personnel with appropriate expertise. A DMR should be performed during the design of new processes and the selection of new equipment. Information from DMRs should be provided to PHA teams for consideration during studies. DMRs should be completed or updated as part of incident investigations and management of change reviews. Key aspects of DMR studies should be the identification of possible damage mechanisms for a process and the determination that materials of construction are resistant to potential damage mechanisms or measures have been taken to prevent or minimize damage from them under process conditions that may be experienced.

3.13. Process safety regulations

Incidents occurred in facilities both covered and not covered by US process safety regulations. However, a case can be made that non-covered facilities departed from good industry practices to a greater extent than for covered facilities. Many facilities that are not covered by process safety regulations in the US pose serious process safety risks, for example, facilities handling reactive chemicals and dusts. These issues have been recognized by the CSB (CSB, 2002, 2006).

In some cases, facilities that are covered by process safety regulations had not implemented a process safety program {24, 32, 34, 36, 48}.

3.14. Compliance with industry and company standards and practices and regulations

Non-compliance with industry and company standards and practices were common in numerous incidents {1, 4, 8, 9, 13, 15, 16, 17, 21, 24, 27, 28, 29, 31, 33, 40, 42, 43, 47, 50, 51, 56, 54b, 54c, 62, 64}. This issue was present for 38% of CSB incidents and 29% of incidents covered by OSHA's PSM standard. Also, there were failures to follow regulations for multiple incidents {8, 12, 16, 24, 25, 33, 34, 36, 39, 40, 41, 64}. This issue was present for 18% of CSB incidents and 14% of incidents covered by OSHA's PSM standard.

Non-compliance with standards, practices and regulations is a fundamental issue for process safety management. If standards, practices and regulations are ignored, the comprehensiveness of their content does not matter and incidents will continue to occur. Certainly, management review and continuous improvement may go some way to address this problem and the inclusion of this CCPS RBPS element (AIChE, 2007b) was addressed in OSHA's RFI (FR, 2013). The RFI also addressed the inclusion of the CCPS RBPS element on measurement and metrics (AIChE, 2007b) to track the effectiveness of process safety management programs. Regulators in California have proposed amendments to the state's process safety regulations for refineries that will require the use of process safety performance indicators (DIR, 2015). Public disclosure of process safety performance would encourage management to pay close attention to ensuring those measures reflect favorably on the company. Thus, for example, it can be expected that there would be fewer action items that remain unclosed past their due dates. Of course, if such an approach were to be applied, it may still be ignored by some companies. Changes in regulatory enforcement may be needed.

A recent case in the UK contrasts markedly with US practices. Phosphine was released from a plant requiring 4500 people to shelter in place for 2–3 h and causing disruption to a motorway (HSE). The U.K. Health and Safety Executive (HSE) investigated the incident and found that an equipment failure caused loss of containment. The company was fined a total of £333,000 and ordered to pay costs of £110,000 after pleading guilty to an offence under the U.K. Health and Safety at Work Act of 1974 having failed to properly assess and act upon the risk of the equipment failing. Such a penalty for an incident of relatively low consequences contrasts with smaller penalties for greater consequence incidents in the United States.

After the prosecution, an HSE inspector said, “This case should serve as a warning to other companies dealing with harmful substances that they need to get their processes absolutely right, in order to ensure the safety of the public, if they don't they will face the consequences.” In virtually, every incident analyzed in this paper, the company did not get it right. There are very few, if any process safety accidents; they are preventable incidents.

3.15. Industry standards and guidelines

Numerous CSB incident investigation reports identified flaws and omissions in industry standards and recommended practices {9, 14, 16, 25, 26, 29, 36, 37, 48, 51, 53, 54c, 55, 56, 63} from organizations such as the American Petroleum Institute (API), the American Society of Mechanical Engineers (ASME), the Compressed Gas Association (CGA), and the National Fire Protection Association (NFPA). Issues with industry standards and guidelines were present for 22% of CSB incidents and 32% of incidents covered by OSHA's PSM standard.

Such standards and practices usually are subject to periodic review and updating as part of a process of continuous improvement so such omissions and deficiencies ultimately can be addressed. However, the CSB has criticized the permissive nature of some of these standards and recommended practices {56} which leave options open to companies not to follow some of the guidance. Also, the CSB has criticized the process by which standards are set by some organizations, for example API {56}.

This finding suggests that CSB should investigate a diversity of incidents to identify as many deficiencies and omissions as possible in industry standards and guidelines.

3.16. Product stewardship

Some incidents involved the use of highly hazardous chemicals obtained by one company from another which did not recognize their hazards {14, 17}. In another case, a company was not informed by a supplier that process safety regulations would need to be implemented before receiving shipments {24}. While chemical supplier companies will not necessarily know the details of how purchasers intend to use the chemicals, they should at least communicate their known hazards to the purchaser. Process safety considerations should be a key part of product stewardship initiatives such as that in the ACC's Responsible Care Code of Management Practices (ACC, 2012). Suppliers of highly hazardous chemicals have responsibilities in supplying such chemicals to other companies.

Some additional issues were identified for a few CSB incidents (see Table 2).

4. Discussion of the analysis

A summary of the results of the analysis is provided in Fig. 2 which shows the percentages of occurrences for each issue. The top issues are abnormal/non-routine work and safeguards which are about equally important for all incidents and for PSM incidents, and stationary source siting for PSM incidents. These are followed by procedures which are about equally important for all incidents and for PSM incidents, and stationary source siting for all incidents. Compliance with standards and practices, maintenance, process changes, previous incidents, and emergency preparedness and response come next. They are followed by issues with design and standards. Compliance with regulations, normalization of deviance, facility siting and audits follow.

Notable differences between the issues for all incidents and PSM incidents occur in some cases. There are significantly more occurrences for PSM incidents than for all incidents for each of stationary source siting, facility siting, and industry standards and guidelines but there are fewer occurrences for PSM incidents than for all incidents for both design and compliance with standards and practices. The difference for stationary source siting is particularly significant.

The lowest percentage of incidents with any of the issues is 13% and the highest is 64%. This is poor process safety performance.

Time trends for the CSB incidents are provided in Table 3. The number of incidents displayed per year in the table largely reflects resources available and decisions made by the CSB. However, it appears that the trend for virtually all issues is for them to continue.

There was an average of 5 issues per incident and as many as 13. Some CSB investigations appear to be more thorough than others so the number of issues identified in the investigation reports does not necessarily reflect all of the issues that were present. Consequently, there may be under counts in the table. However, it is clear that most incidents involved multiple issues and this conclusion would not change. It is likely that the underlying reason for multiple issues being present for the majority of incidents is a systemic problem with the process safety management system and process safety culture.

Table 2
Additional issues noted for CSB incidents.

Issue	Examples
Process boundaries defined too narrowly	59
Inadequate workforce participation	56
Inadequate process safety information	3, 6
Inaccurate or incomplete safety data sheets	2, 5, 15, 38
Loss of process design and safety information due to poor management of records	11
Lack of hazard recognition and understanding	14, 15, 21, 26, 31
Inadequate communication of plant hazards	5, 17
Need for positive materials identification during commissioning, modifications, and maintenance	10
Inadequate or no pre-startup safety review	3, 13, 29, 45
No or inadequate near miss reporting	54c
Inadequate or no incident investigation	12, 20
Inadequate contractor management	39, 59, 64
Outstanding action items	2, 4, 6, 7, 16, 25, 45, 54b, 54c, 63
Inadequate oversight by management	4, 7, 10, 12, 29, 32

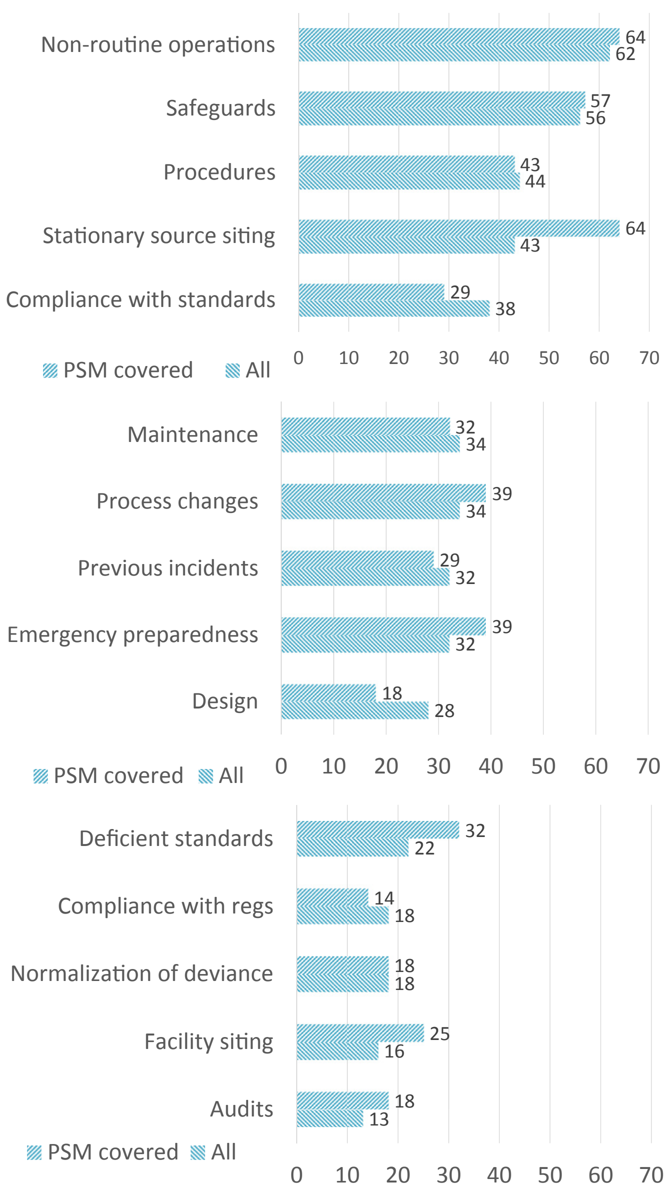

Fig. 2. Analysis of CSB incidents.

5. Further observations

This section provides some additional noteworthy observations that were made during the review of incident descriptions.

5.1. Age of plants

Many process safety incidents have been attributed to aging plants (HSE, 2010). Of course, the longer a process operates the more exposure there is to risk. Plant aging factors need to be addressed by process safety programs.

Equipment deteriorates over time and this has played a role in some incidents {63}. Damage mechanism reviews can help to address this issue. Old equipment may be used, for example, a tank constructed in 1929 failed catastrophically in 2008 {47}. Use of old equipment must be subject to close review.

As time passes, industry standards and best practices evolve but the designs of processes are frozen in time. Thus, gaps between current industry standards and best practices and existing company standards and practices may exist. Use of dated standards and practices has played a role in several incidents {30}. Outdated technology also has played a role in some incidents {29}. Process safety programs should require periodic formal reviews to identify and evaluate the feasibility of adopting current standards, best practices, and technology retrospectively. Current standards, best practices, and technology should be addressed when developing recommendations to reduce risk, for example, in PHA studies, MOC reviews, and incident investigations. OSHA's PSM standard does not require employers to evaluate updates to applicable RAGAGEP or to examine new RAGAGEP after evaluating and documenting compliance with mechanical integrity requirements. OSHA's RFI sought comments on the best approach for requiring the evaluation of updates to applicable RAGAGEP. However, RAGAGEP is defined narrowly by CCPS which is the definition being advanced by OSHA. It covers generally approved ways to perform specific engineering, inspection, or mechanical integrity activities, such as fabricating a vessel, inspecting a storage tank, or servicing a relief valve. There are many more types of standards and practices that could be included in the definition of RAGAGEP.

Informal procedures may become institutionalized with the passage of time with operators passing them on verbally without others being aware {46}. Key tribal knowledge may be lost through departures of personnel including retirements {11, 21, 54}. An aging workforce may display diminishing physical and mental faculties. These are human factors issues that should be addressed by a human factors program.

Often process facilities have been built in rural areas away from

Table 3
Time trends for CSB incidents.

Issue	1998		1999		2000		2001		2002		2003		2004		2005		2006		2007		2008		2009		2010		2011		2012	
	N	P	N	P	N	P	N	P	N	P	N	P	N	P	N	P	N	P	N	P	N	P	N	P	N	P	N	P	N	P
Design	1	–	–	1	–	–	1	–	–	–	5	–	–	1	–	1	1	1	2	1	1	–	–	–	2	–	–	–	1	–
Safeguards	1	–	–	–	–	–	1	–	4	–	4	2	–	3	–	3	1	3	2	1	1	1	2	2	3	1	1	–	1	–
Procedures	2	–	–	1	–	–	1	–	3	–	5	2	–	1	–	2	1	1	1	–	2	2	1	–	–	2	2	–	–	–
Maintenance	–	–	–	1	–	–	2	–	2	1	3	–	–	1	–	1	1	1	1	–	1	–	2	1	–	2	2	–	–	1
Normalization of deviance	–	–	–	–	–	–	–	–	1	–	2	1	1	–	–	1	–	–	–	–	1	1	–	–	1	–	1	–	–	1
Abnormal and non-routine operations	1	–	–	2	–	–	3	–	1	1	4	2	1	3	–	1	2	1	2	–	2	3	1	1	4	3	2	–	1	1
Process changes	1	–	–	1	–	–	2	–	1	–	5	–	1	1	–	2	–	1	–	1	–	2	–	–	1	3	–	–	–	1
Previous incidents	1	–	–	–	–	–	2	–	1	–	2	2	–	2	–	1	–	–	1	–	1	1	2	–	2	1	1	–	1	1
Stationary source siting	1	–	–	1	–	–	–	–	1	1	4	3	1	4	–	2	1	2	1	–	–	2	2	2	–	–	–	–	–	1
Facility siting	–	–	–	–	–	–	–	–	1	–	1	2	–	2	–	1	–	1	1	–	–	–	–	1	–	–	1	–	–	–
Emergency preparedness and response	–	–	–	–	–	–	1	–	2	1	1	3	–	2	–	–	1	2	1	–	1	2	1	–	2	–	–	–	1	1
Audits and inspections	–	–	–	1	–	–	–	–	–	–	2	1	–	–	–	1	–	–	–	–	1	–	–	–	–	2	–	–	1	–
Compliance with industry and company standards and practices	1	–	–	1	–	–	–	–	2	–	5	–	1	1	–	2	2	–	1	–	2	1	1	1	1	2	1	–	1	–
Failure to follow regulations	–	–	–	–	–	–	–	–	2	–	1	–	–	2	–	–	1	2	3	–	–	–	–	–	–	–	–	–	1	–
Industry standards and guidelines	–	–	–	–	–	–	–	–	1	–	2	–	–	2	–	1	–	1	–	1	–	–	2	1	1	2	–	–	–	1

N –Not covered by PSM. P – Covered by PSM.

the public but over time housing developments are built near the facility. This issue was present for several incidents. It must be addressed through land use planning.

5.2. Magnitude of consequences

The consequences of some incidents could have been much worse. For example, releases occurred under favorable weather conditions {22}, with the wind blowing away from populated areas {10}, when projectiles created by explosions missed hitting adjacent processes {11}, and at a time when the number of people potentially exposed was at a minimum {3, 36}. In some cases, people exposed to serious hazards miraculously escaped harm {63} and avoided being caught in the collapse of structures {49}.

These situations were serendipitous. However, it does suggest a hazard control strategy in which riskier activities, such as non-routine work, should be timed for when the risk to receptors is at a minimum. Also, facility personnel in the vicinity of a riskier activity who are not part of the activity should be relocated for the duration of the activity. Such considerations should be an integral part of planning for non-routine work.

Some injuries occurred when people investigated process abnormalities {32, 45, 64} or tried to fix leaks {62, 63}. Such situations must be taken seriously and not considered part of normal operations. Appropriate precautions must be taken.

6. Underlying issues

This section identifies three issues that underlie all of the other issues. Each underlying issue is discussed and recommendations are made for addressing it.

6.1. Competency of facility personnel

The CSB has not explicitly addressed the issue of competency in many incident investigations. However, it was mentioned as a contributor in at least one CSB incident {29}. In reading incident descriptions it is hard not to reach the conclusion that the people responsible for the processes were not competent in one or more ways in many cases. This issue likely underlies most of the process safety incidents that have occurred.

The decisions made by personnel ranging from managers to operators can be questioned for many incidents. While facility personnel were not obviously incompetent, in numerous cases clearly they were not competent to perform the work required of them safely. In particular, facility personnel often lacked an awareness and/or understanding of one or more critical aspects of a process, such as its hazards {6, 8, 11, 24, 32, 36, 39, 41, 44, 54a, 54c, 62}. Inadequate operator and mechanic training was an issue in a number of incidents {1, 2, 3, 10, 11, 12, 13, 14, 18, 21, 22, 25, 28, 29, 32, 39, 42, 45, 64}.

OSHA's RFI sought comments on the addition of an element on process safety competency (FR, 2013) as described by CCPS (AIChE, 2007b). Certainly, facilities should implement a competency management program for process safety but the CCPS RBPS competency approach should be modified (Baybutt). Critical personnel should be certified and periodically re-certified. For example, operators and mechanics should not be allowed to operate or maintain a process without demonstrating their competence, including such topics as knowledge and understanding of process hazards, the higher risks of non-steady state operations such as startup, previous incidents, the safe operating envelope for the process, handling abnormal and emergency situations, conditions under which the process must be shut down, use of permit-to-work systems, and the identification and management of changes. Responsibility for supervision of the operation of a process should be formally assigned to a single person and the responsibility formally transferred to another individual as necessary. There must be supervisory oversight of processes at all times.

In some incidents, the availability of technical expertise has been an issue {29, 32, 45}. When critical decisions are faced regarding process safety, someone must be present who understands mass and energy balances, physical and chemical phenomena that may occur, process controls and safeguards, hazards present, and the safe operating envelope, that is, tolerable ranges of values of safety-critical parameters. Individuals should be designated as engineering and technical authorities who ensure the proper application of engineering standards and technical practices and adjudicate when deviations are proposed. They must be available for consultation at all times and they must have the authority to shut down processes without the need for consultation with managers and without recrimination. They must be free of production or commercial pressures and should report directly to upper management. A stop work authority that authorizes personnel to stop work in the face of imminent risk or dangerous activity and an ultimate work authority that requires designation of who has the authority for operational safety and decision making are addressed by OSHA's RFI (FR, 2013).

6.2. Standards for process safety elements

Process safety programs comprise multiple elements such as employee participation, PSI, PHA, etc. For example, OSHA's PSM standard has 14 elements (CFR) and CCPS's RBPS program has 20 elements (AIChE, 2007b). However, it is possible for companies to address each of these elements in widely varying ways because their requirements are general in nature and do not address the effectiveness of their contribution to risk reduction. For example, PHA studies can be conducted with widely varying levels of quality. A solution to this problem is to establish minimum performance standards for each element. These standards should be established by an international organization such as ISO so they apply around the world. There is precedent for such standards in the IEC 61511 standard for safety instrumented systems (ANSI/ISA, 2004a, b, c) and the Hazard and Operability Study (HAZOP) standard (IEC, 2001), although the latter is outdated. These standards are still performance-based in nature, as should be new standards for other process safety elements.

One benefit of standards for process safety elements is that they can be updated more easily than regulations. Current US process safety regulations are static and unable to adapt to innovation and advances in the management of major hazard risks.

This problem could be overcome if the US process safety regulations simply established overall requirements for each process safety element and required companies to follow RAGAGEP for each element. There is already a precedent for doing so with the PSI and mechanical integrity PSM elements although RAGAGEP is interpreted more narrowly in that context. The proposal is to broaden the meaning of RAGAGEP, establish RAGAGEPs for all process safety elements, and require that RAGAGEPs be followed. The process safety regulations would effectively update automatically as RAGAGEPs evolve. Continuous improvement is an essential element of a management system for any surety such as quality, environmental protection, and occupational health and safety, and is part of CCPS's RBPS program (AIChE, 2007b).

6.3. Safety goals and risk tolerance criteria

In at least one incident, the CSB concluded that one company had knowledge of and acceptance of high risk {29}. Most companies do not really know what risk their process facilities pose because they do not evaluate risk as there are no requirements to do so under the current US process safety regulatory regime. Indeed, it is possible that similar facilities may pose quite different risks because the regulations do not contain any safety goals or risk tolerance criteria.

The goal of process safety programs is to ensure safety and control risks. Consequently, some form of safety goals or risk tolerance criteria are needed (Baybutt, 2015c). Measures are needed to determine achievement of goals and criteria. In the absence of process safety goals and measures to determine their achievement, resources may be allocated inappropriately. There is a current precedent for the use of numerical risk tolerance criteria provided by the New Jersey Toxic Catastrophe Prevention Act (TCPA) process safety regulations (NJDEP). Although the particular approach used is erroneous (Baybutt, 2014a), companies in the state have accepted the use of numerical criteria.

Many jurisdictions around the world employ the tolerability of risk framework developed by the UK HSE including the As Low As Reasonably Practicable (ALARP) principle (Baybutt, 2014b). Its use should be incorporated into US regulations. Although the ALARP principle usually incorporates the use of numerical criteria, it can be used without specifying such criteria but using its essential element that risk should be reduced until the incremental expenditures of resources is grossly disproportionate to the value of the incremental risk reduction achieved. This is a performance-based requirement that is consistent with the current philosophy of US process safety regulations. Use of the ALARP principle would help to accommodate improved process safety practices without the need for periodic modification of regulations.

One of the difficulties faced by PHA teams is the use of risk ranking in studies to determine if recommendations for risk reduction, that is, additional safeguards, are needed. The CSB has been critical of the subjectivity involved and the criticism is warranted (Baybutt, 2013a). The CSB has suggested that Layers of Protection Analysis (LOPA) be used for this purpose. Of course, LOPA requires the use of risk tolerance criteria that are allocated from facility criteria (Baybutt, 2014c). Proposed amendments to the California process safety regulations include requirements for such an analysis (DIR, 2015). Simply put, the adequacy of safeguards cannot be determined without risk tolerance criteria.

Use of risk tolerance criteria with LOPA raises the issue of what type of risk is being addressed and should be addressed. Certainly, no loss of life is acceptable but in US society single fatalities are more tolerable than multiple fatalities as evidenced by the relative attention paid by the media to single versus multiple fatality incidents. Perhaps the focus in process safety should be on protecting groups of people, particularly members of the public.

7. Recommendations

Several recommendations can be made to strengthen process safety programs.

7.1. Recommendations for operating companies

- Add process safety elements on:
  - Design and engineering
  - Human factors review
  - Facility and stationary source siting review
  - Safeguards protection analysis
  - Operational discipline
  - Competency management
  - Previous incidents review
  - Best practices review
- Address non-routine operations throughout all process safety elements.
- Require the use of job hazard analysis for non-routine work.
- Expand the types of changes addressed under the MOC element to include any change that may impact on or affect a release of a highly hazardous chemical.
- Make compliance audits more comprehensive.
- Require annual emergency response drills for facilities.
- Require that process risks be managed using the ALARP principle.
- Require facilities to set safety goals and/or risk tolerance criteria and demonstrate compliance with them.

7.2. Recommendations for standards organizations

- Develop international standards for key process safety elements that can be referenced by process safety regulations.

7.3. Recommendations for the Chemical Safety Board

- Expand the incident attributes that are addressed in investigations to gain more insights into process safety

performance, for example, explicitly address competency, human and organizational factors, safety culture, plant aging factors, and operational discipline.
- Provide an appendix to each investigation report that characterizes each attribute for the incident. Not only is information on deficiencies and omissions valuable but also knowledge of what was done correctly or that there were no problems in certain areas is valuable.
- Compile a data base of incidents and their attributes that is searchable.
- Provide periodic analyses of the full data base of incidents looking for commonalities and trends and make recommendations for improvements in process safety practices.
- Publish the criteria used for the selection of incidents that are investigated by the CSB and update as appropriate.
- Expand on the current documentation in incident investigation reports to explain how the criteria were applied in selecting the incident for investigation.

8. Conclusions

Process safety programs aim to identify possible incidents before they occur and implement practices to prevent them. Remarkably, all of the reviewed incidents involved some type of deficiency or omission in adhering to established process safety practices. In many cases there were multiple deficiencies and omissions. Therefore, it should have come as no surprise that these processes experienced incidents. All of the incidents were predictable and preventable. Just because an event may be described as unthinkable does not necessarily mean it is unpredictable.

The quote, “It is said that there is nothing new in safety - it has all happened before” was published by CCPS [1]. Virtually every incident or something like it had happened somewhere before.

This survey of incidents covered a range of company and facility sizes, and types of processes. Also, it included facilities covered and not covered by process safety regulations. Thus, the results of the analysis apply to diverse processes. They can be used by companies to focus on improving areas where other companies have fallen short.

The CSB adopted a motto in its early days; it still applies: Salus populi suprema lex esto (People’s safety should be the highest law).

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